Program : **Diploma in Polymer Technology**	
Course Code : **5092**	Course Title: **Quality Assurance &Testing**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To understand the significance in quality control of polymer products.
- To impart knowledge on raw material testing, processability tests and product testing which comes under both rubbers and plastics.

Course Prerequisites:

Topic	Course code	Course name	Semester
Quality, ISO		Industrial management and safety	5
Purity tests of DPG, MBT, ZnO Melting point		Polymer science Lab	3
Tensile strength, M_{300}, EB		Physical testing lab	3
Specification tests on dry rubber		QA and Testing lab	5
Specification tests on NR Latex		Latex technology lab	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Generalise various standard organizations, thermal and, electrical properties of polymers.	15	Understanding
CO2	Summarize test methods for ISNR and concentrated latex as per IS standards	14	Understanding

CO3	Explain the testing procedures for physical properties of polymers.	15	Understanding
CO4	Explain the specific testing procedures for dry rubber, plastic and latex products	14	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	**Generalise various standard organizations, thermal and, electrical properties of polymers**		
M1.01	Outline the quality concepts and importance of standards organizations	1	Remembering
M1.02	Summarize the role of standards organizations in standardization and quality assurance.	3	Understanding
M1.03	Explain test procedures for purity of raw materials	4	Understanding
M1.04	Describe test methods for thermal and electrical properties of polymers.	7	Understanding

Contents:

Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction.

Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO.

Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT.

Electrical properties - Dielectric strength, dielectric constant, arc resistance.

CO2	**Summarize test methods for ISNR and concentrated latex as per IS standards**		
M2.01	Explain the significance, principle and procedure to determine tests of dry rubber (ISNR).	6	Understanding
M2.02	Compile the technical specifications of ISNR as per IS 4588.	1	Understanding
M2.03	Explain the significance, principle and procedure to determine tests of concentrated latex.	6	Understanding
M2.04	Compile the technical specifications of concentrated latex as per IS 3708.	1	Understanding
	Series Test I	1	

Contents:

Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N_2 content, P_o and PRI, ASH and Colour. Technical specifications of ISNR as per BIS.

Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity

CO3	**Explain the testing procedures for physical properties of polymers**		
M3.01	Describe the test method to determine tensile properties of plastics and rubbers	4	Understanding
M3.02	Describe the test method to determine permanent set of rubbers.	4	Understanding
M 3.03	Describe the test method to determine hardness, abrasion resistance, resilience, flex crack, fatigue failure and impact testing.	6	Understanding
M3.04	Outline the procedure to determine MFI of plastics.	1	Understanding

Contents:

Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers – Tensile strength, M_{300}, Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy

Processability tests -Determination of melt flow index of plastics.

CO4	**Explain the specific testing procedures for dry rubber, plastic and latex products**

M4.01	State the importance of product testing.	1	Remembering
M4.02	Outline the test procedure and specifications for dry rubber products.	5	Understanding
M4.03	Explain the test procedure and specifications for latex products.	4	Understanding
M4.04	Explain the test procedure and specifications for plastic products.	4	Understanding
	Series Test II	1	

Contents:

Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps.

Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves.

Testing of Plastic product - PVC pipe, Water tank, Helmet

Text / Reference:

T/R	Book Title/Author
T1	Physical testing of Rubbers – Scott
R1	Physical testing of Rubbers – R.P. Brown
R2	Indian Standards
R3	ASTM standards – Vol. 8 & 9 series
R4	Plastic Testing Technology – Vishu Shah

Online Resources:

Sl.No	Website Link
1	https://www.astm.org
2	https://www.astm.org
3	https://link.springer.com
4	http://www.intertek.com
5	https://www.smithersrapra.com